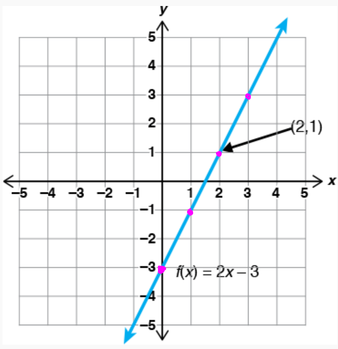
Understanding a Graph of a Function

Like an equation, a function can be graphed on a coordinate grid.

Example 1

Hide

For $f(x) = 2x - 3$, find $f(2)$.



Straight line with slope = 2 and y-int. of −3

Read the graph to find $f(2)$. What is the y-coordinate of the line when the x-coordinate is 2?
The y-coordinate is 1, so $\boldsymbol{f(2) = 1}$.
The answer makes sense.
Try substituting 2 for x.
$f(2) = 2(2) - 3$
$= 4 - 3$
$= 1$

The answer is the same.

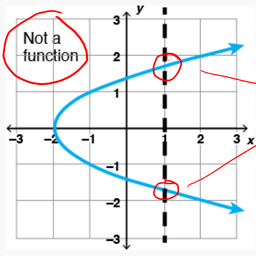
A graph can also help you decide whether a relation is a function. Remember, if a relation is a function, every input can have only one output. That means that a graph of a function will have only one y-coordinate for every x-coordinate.

The **vertical line test** is a quick way to judge whether a graph represents a function. If you can draw a vertical line that passes through more than one point anywhere on a graph, the graph is not a function.

Example 2

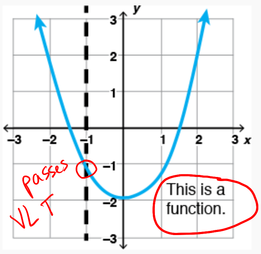
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On each graph, the blue line represents a relation. Is the relation a function?
Imagine drawing a vertical line through each graph.
If the vertical line passes through the graph more than once, then there is an x-coordinate with more than one possible y-coordinate.



fails VLT (vertical line test)!

The vertical line crosses the blue line more than once.

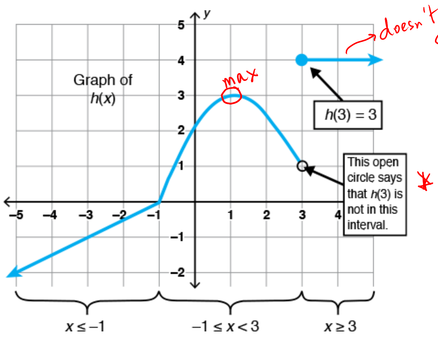


No matter where you place the vertical line, it will cross the blue line only once.

You may be asked a question about a specific interval on a graph. An **interval** is the distance between two points, usually on the x-axis.

The graph of function $h(x)$ has several parts. Notice how intervals can be used to describe and analyze the graph.

- During the interval $x \le -1$, $h(x)$ increases as x increases.
- In the interval $-1 \le x < 3$, the function reaches its highest point when $x = 1$.
- The function does not change in the interval $x \ge 3$.



doesn't change

max

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